

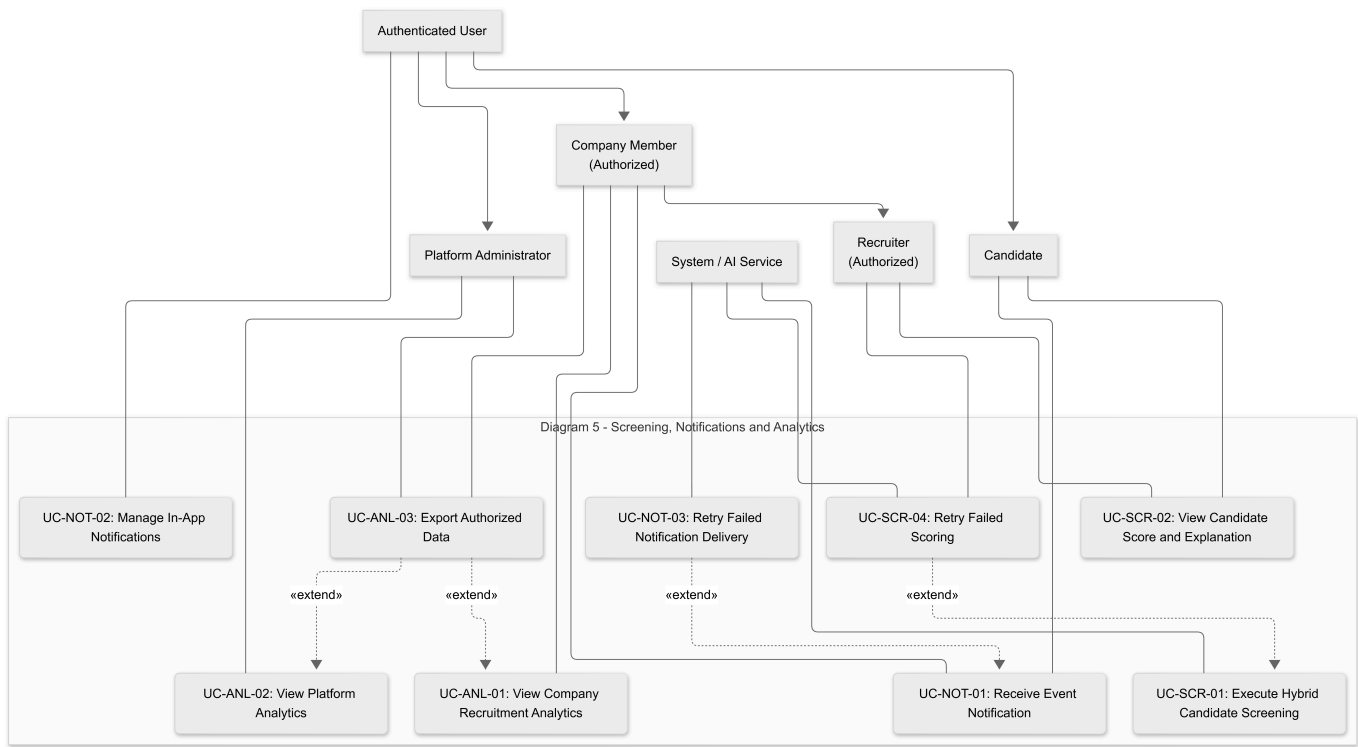
Use Case Specifications: Diagram 5 - Supporting Services and Analytics

Performed by: Lưu Chí Hải | Reviewed by: Nguyễn Gia Quốc Uy | Edited by: Lưu Chí Hải

1. Use Case Diagram

Performed by: Lưu Chí Hải | Reviewed by: Nguyễn Gia Quốc Uy | Edited by: Lưu Chí Hải

Below is the static render of the Diagram for PDF. The Mermaid source code is attached underneath for reference.



2. Mermaid Source Code

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```
---
config:
  theme: neutral
  flowchart:
    defaultRenderer: elk
---
flowchart TB
  %% Actors
  sys["System / AI Service"]
  auth_user["Authenticated User"]
  cand["Candidate"]
  cm["Company Member\n(Authorized)"]
  rec["Recruiter\n(Authorized)"]
  admin["Platform Administrator"]
```

```

%% Actor Generalization
auth_user --> cand
auth_user --> cm
auth_user --> admin
cm --> rec

%% System Boundary
subgraph subGraph0["Diagram 5 - Screening, Notifications and Analytics"]
    direction TB
    UC_SCR_01("UC-SCR-01: Execute Hybrid Candidate Screening")
    UC_SCR_02("UC-SCR-02: View Candidate Score and Explanation")
    UC_SCR_04("UC-SCR-04: Retry Failed Scoring")
    UC_NOT_01("UC-NOT-01: Receive Event Notification")
    UC_NOT_02("UC-NOT-02: Manage In-App Notifications")
    UC_NOT_03("UC-NOT-03: Retry Failed Notification Delivery")
    UC_ANL_01("UC-ANL-01: View Company Recruitment Analytics")
    UC_ANL_02("UC-ANL-02: View Platform Analytics")
    UC_ANL_03("UC-ANL-03: Export Authorized Data")
end

%% Actor to Use Case Relationships
sys --- UC_SCR_01
sys --- UC_SCR_04
sys --- UC_NOT_03

rec --- UC_SCR_02
rec --- UC_SCR_04

cand --- UC_SCR_02
cand --- UC_NOT_01

cm --- UC_NOT_01
cm --- UC_ANL_01
cm --- UC_ANL_03

auth_user --- UC_NOT_02

admin --- UC_ANL_02
admin --- UC_ANL_03

%% Use Case to Use Case Relationships (Extend)
UC_SCR_04 -. "«extend»" .-> UC_SCR_01
UC_NOT_03 -. "«extend»" .-> UC_NOT_01
UC_ANL_03 -. "«extend»" .-> UC_ANL_01
UC_ANL_03 -. "«extend»" .-> UC_ANL_02

```

3. Traceability Summary

Use Case ID	Use Case Name	Actor(s)	Covered Requirements
UC-SCR-01	Execute Hybrid Candidate Screening	System / AI Service	FR-SCR-01 to FR-SCR-08
UC-SCR-02	View Candidate Score and Explanation	Recruiter, Candidate	FR-SCR-08, FR-SCR-09, FR-SCR-11, FR-SCR-13
UC-SCR-04	Retry Failed Scoring	Authorized Recruiter, System	FR-SCR-12
UC-NOT-01	Receive Event Notification	Candidate, Company Member	FR-NOT-01 to FR-NOT-03
UC-NOT-02	Manage In-App Notifications	Authenticated User	FR-NOT-04 to FR-NOT-06
UC-NOT-03	Retry Failed Notification Delivery	System	FR-NOT-07, FR-NOT-08
UC-ANL-01	View Company Recruitment Analytics	Authorized Company Member	FR-ANL-01, FR-ANL-02, FR-ANL-04, FR-ANL-07
UC-ANL-02	View Platform Analytics	Platform Administrator	FR-ANL-03, FR-ANL-04, FR-ANL-07
UC-ANL-03	Export Authorized Data	Authorized User(Company Member, Platform Administrator)	FR-ANL-05 to FR-ANL-07

4. Use Case Specifications

4.1. UC-SCR-01: Execute Hybrid Candidate Screening

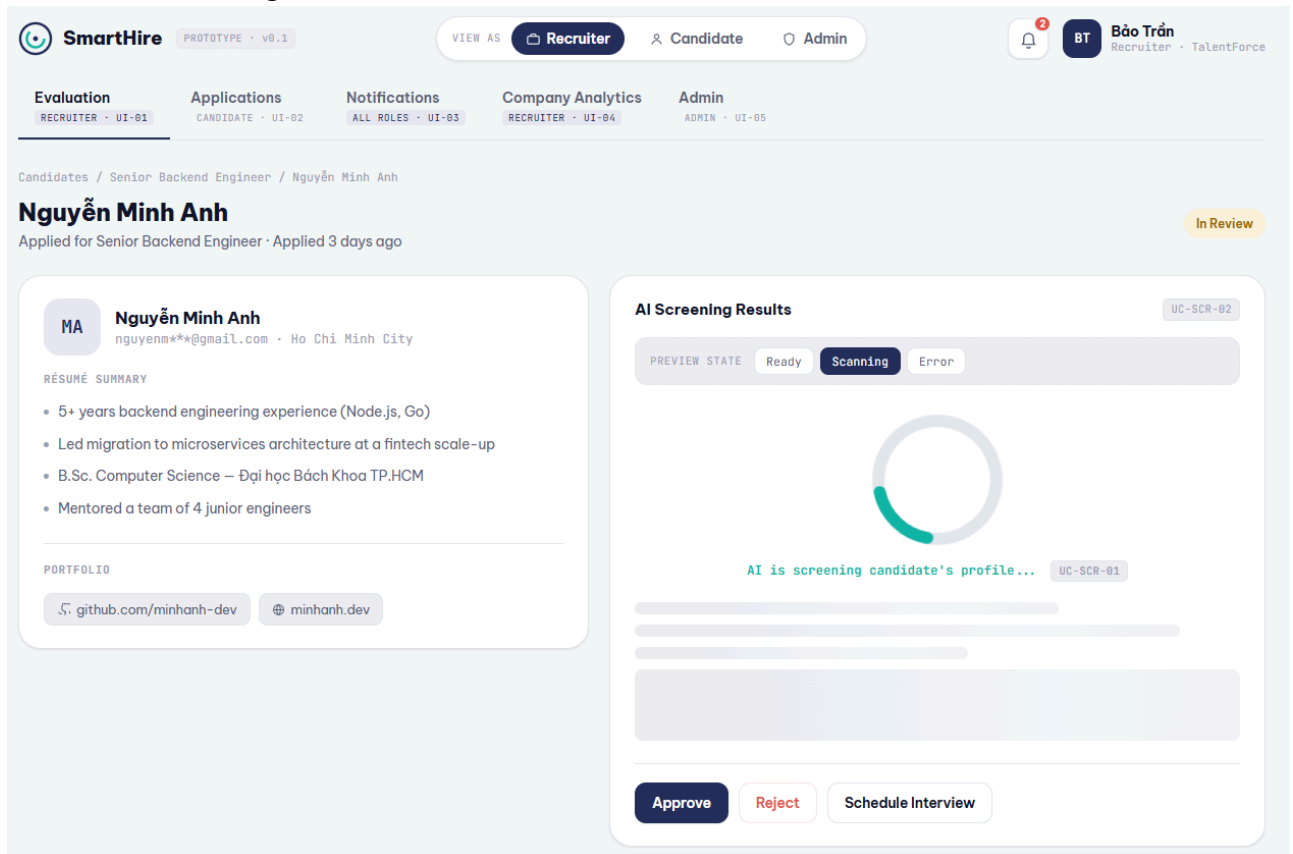
- **Use-case ID:** UC-SCR-01
- **Use-case Name:** Execute Hybrid Candidate Screening
- **Actor(s):** System / AI Service
- **Description:** The system automatically initiates and executes a hybrid evaluation process (deterministic matching and AI semantic evaluation) asynchronously after a candidate successfully submits a job application.
- **Preconditions:** A valid job application has been successfully submitted and normalized CV data is available.
- **Basic Flow:**
 1. **System** detects a newly submitted application.
 2. **System** updates the screening status to **Processing**.
 3. **System** calculates the deterministic matching score based on structured requirements.
 4. **System** triggers the **AI Service** to perform semantic evaluation on the candidate's CV.
 5. **AI Service** returns the AI score and a human-readable explanation.
 6. **System** calculates the final blended score and classification (High/Moderate/Low Match).
 7. **System** updates the application screening status to **Completed**.
- **Alternative Flows:**

- **AF-01: AI Scoring Failure (at Step 5):**

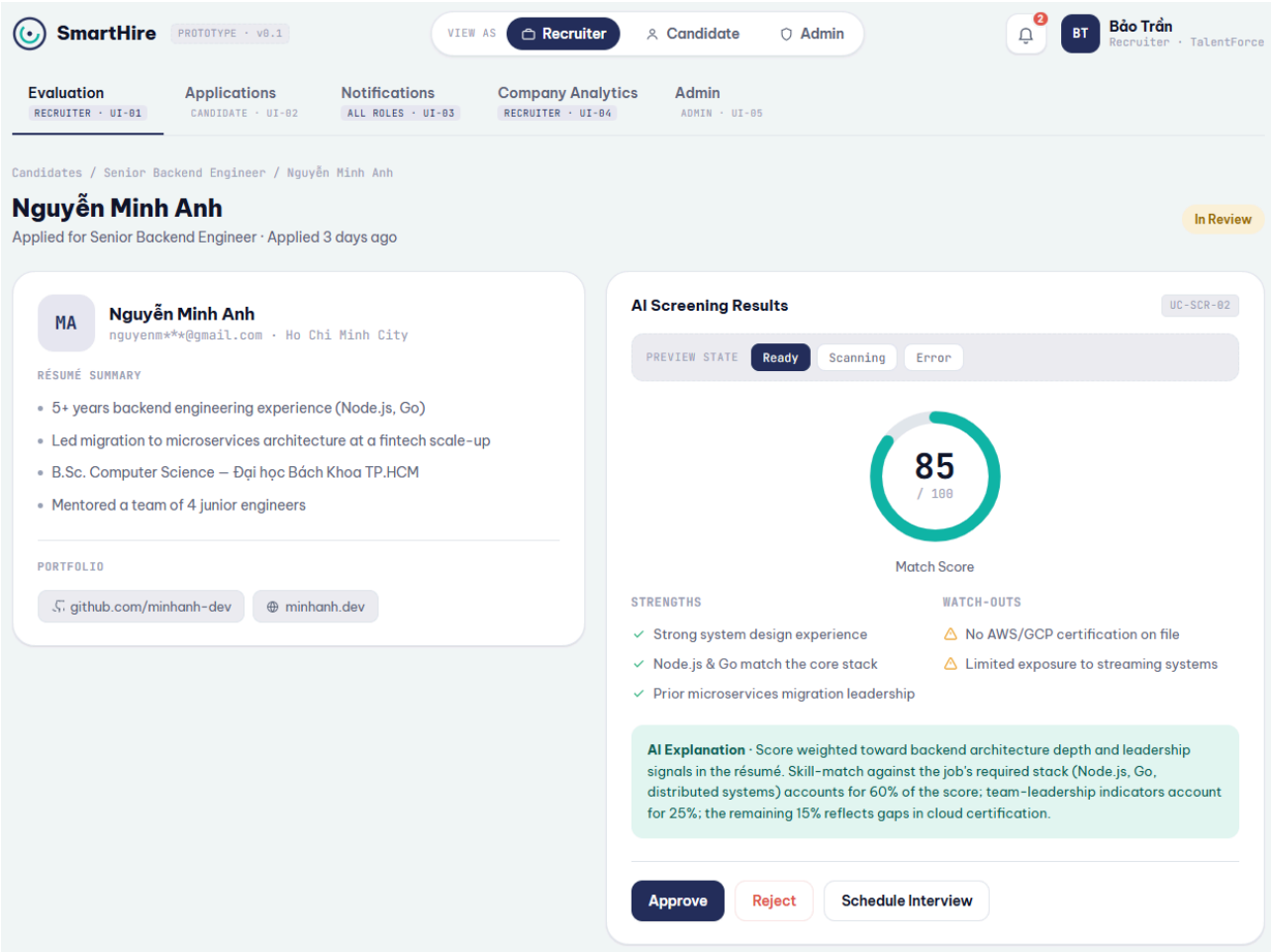
1. If the **AI Service** fails to respond or returns an error, the **System** sets the screening status to **Failed**.
 2. **System** logs the error for a manual retry (via UC-SCR-04).
- **Postconditions:** The application has a finalized hybrid score or is marked as failed.
 - **Special Requirements:** Processing time should ideally not exceed 30 seconds to maintain real-time responsiveness for recruiters viewing newly submitted applications.

Prototype Evidence:

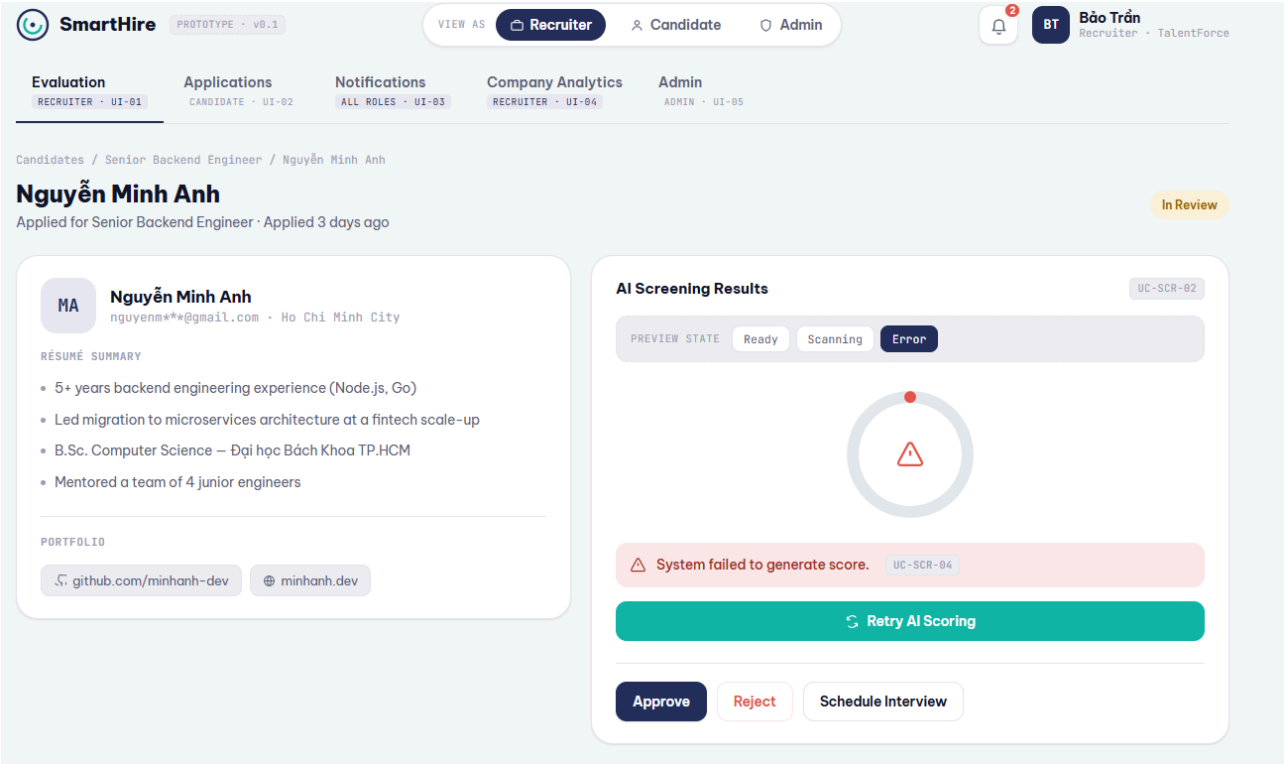
- Basic Flow (Processing):



• Basic Flow (Completed):



• AF-01 (Failed):



4.2. UC-SCR-02: View Candidate Score and Explanation

• Use-case ID: UC-SCR-02

- **Use-case Name:** View Candidate Score and Explanation
- **Actor(s):** Recruiter (Authorized), Candidate
- **Description:** Allows an authorized recruiter to view the full screening score and AI-generated explanation, while allowing the candidate to view limited, non-confidential screening progress.
- **Preconditions:** The screening process (UC-SCR-01) has been initiated or completed. The actor is authenticated.
- **Basic Flow (Recruiter View):**
 1. **Recruiter** navigates to the application detail view for a specific candidate.
 2. **System** verifies company ownership and permissions.
 3. **System** retrieves and displays the detailed hybrid score, gauge chart, and AI explanation (Strengths/Watch-outs).
- **Alternative Flows:**
 - **AF-01: Candidate View (at Step 2):**
 1. If the actor is the **Candidate**, the **System** hides the detailed score ring and AI notes.
 2. **System** displays a general, candidate-friendly status (e.g., "Your application is being evaluated").
- **Postconditions:** The actor views the score details corresponding to their role permissions.
- **Special Requirements:** None.

Prototype Evidence:

- Basic Flow (Recruiter):

The screenshot displays the SmartHire application interface. At the top, there's a navigation bar with the SmartHire logo, a version indicator (v0.1), and a 'VIEW AS' dropdown menu currently set to 'Recruiter'. Other menu items include 'Candidate' and 'Admin'. A user profile for 'Bảo Trần' (Recruiter, TalentForce) is visible in the top right corner.

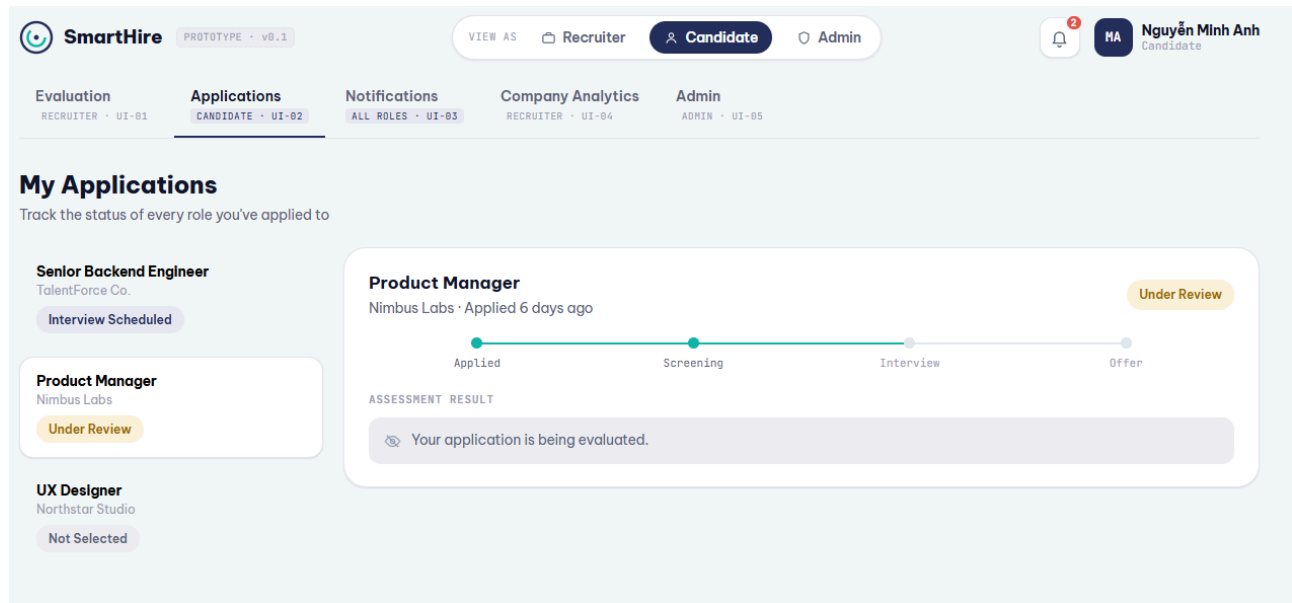
The main content area shows the 'Evaluation' tab selected, with sub-tabs for 'RECRUITER · UI-01', 'Applications · CANDIDATE · UI-02', 'Notifications · ALL ROLES · UI-03', 'Company Analytics · RECRUITER · UI-04', and 'Admin · ADMIN · UI-05'. The breadcrumb trail indicates the path: 'Candidates / Senior Backend Engineer / Nguyễn Minh Anh'.

The candidate's name 'Nguyễn Minh Anh' is prominently displayed, along with the status 'Applied for Senior Backend Engineer · Applied 3 days ago' and an 'In Review' badge. The candidate's profile includes a 'RÉSUMÉ SUMMARY' with bullet points: '5+ years backend engineering experience (Node.js, Go)', 'Led migration to microservices architecture at a fintech scale-up', 'B.Sc. Computer Science – Đại học Bách Khoa TP.HCM', and 'Mentored a team of 4 junior engineers'. A 'PORTFOLIO' section shows links to 'github.com/minhanh-dev' and 'minhanh.dev'.

The 'AI Screening Results' section (UC-SCR-02) features a 'PREVIEW STATE' dropdown set to 'Ready'. A large circular gauge shows a 'Match Score' of 85 out of 100. Below this, 'STRENGTHS' are listed: 'Strong system design experience', 'Node.js & Go match the core stack', and 'Prior microservices migration leadership'. 'WATCH-OUTS' include 'No AWS/GCP certification on file' and 'Limited exposure to streaming systems'. An 'AI Explanation' box states: 'Score weighted toward backend architecture depth and leadership signals in the résumé. Skill-match against the job's required stack (Node.js, Go, distributed systems) accounts for 60% of the score; team-leadership indicators account for 25%; the remaining 15% reflects gaps in cloud certification.'

At the bottom, there are three action buttons: 'Approve', 'Reject', and 'Schedule Interview'.

- AF-01 (Candidate View):

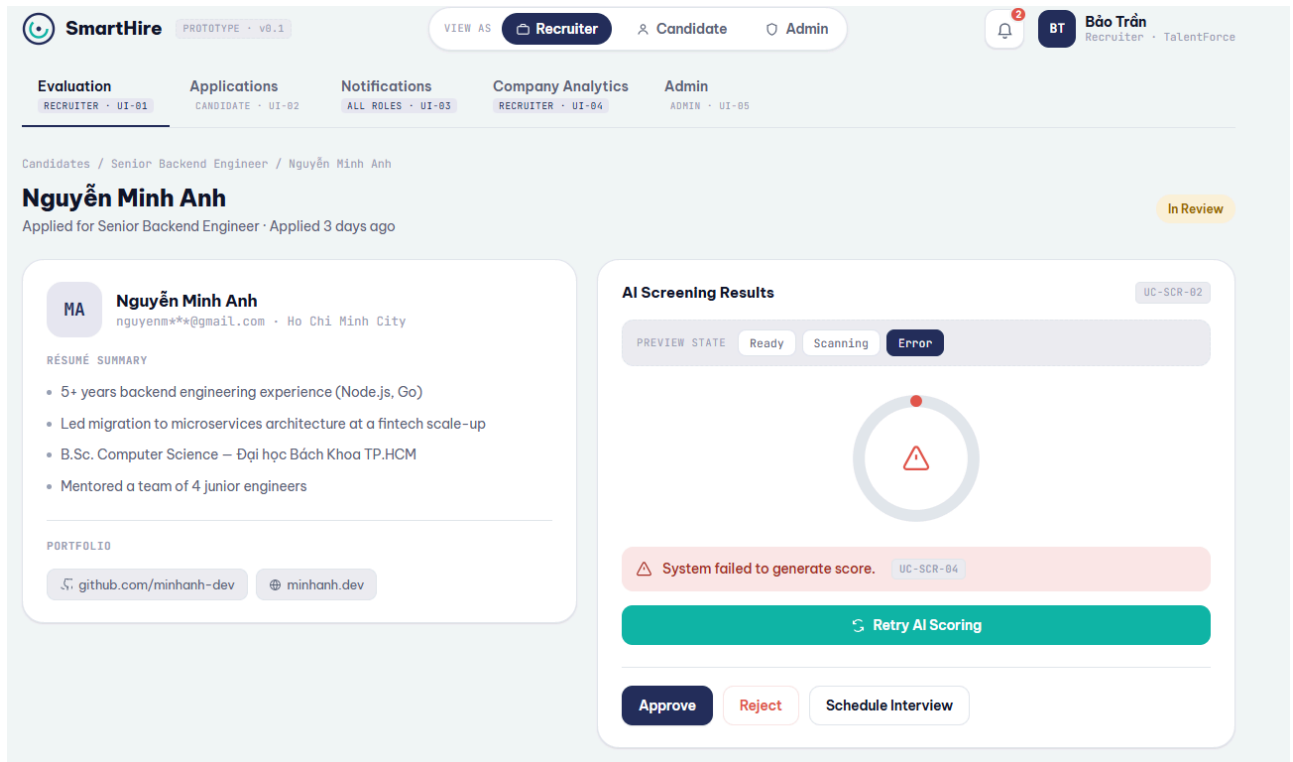


4.3. UC-SCR-04: Retry Failed Scoring

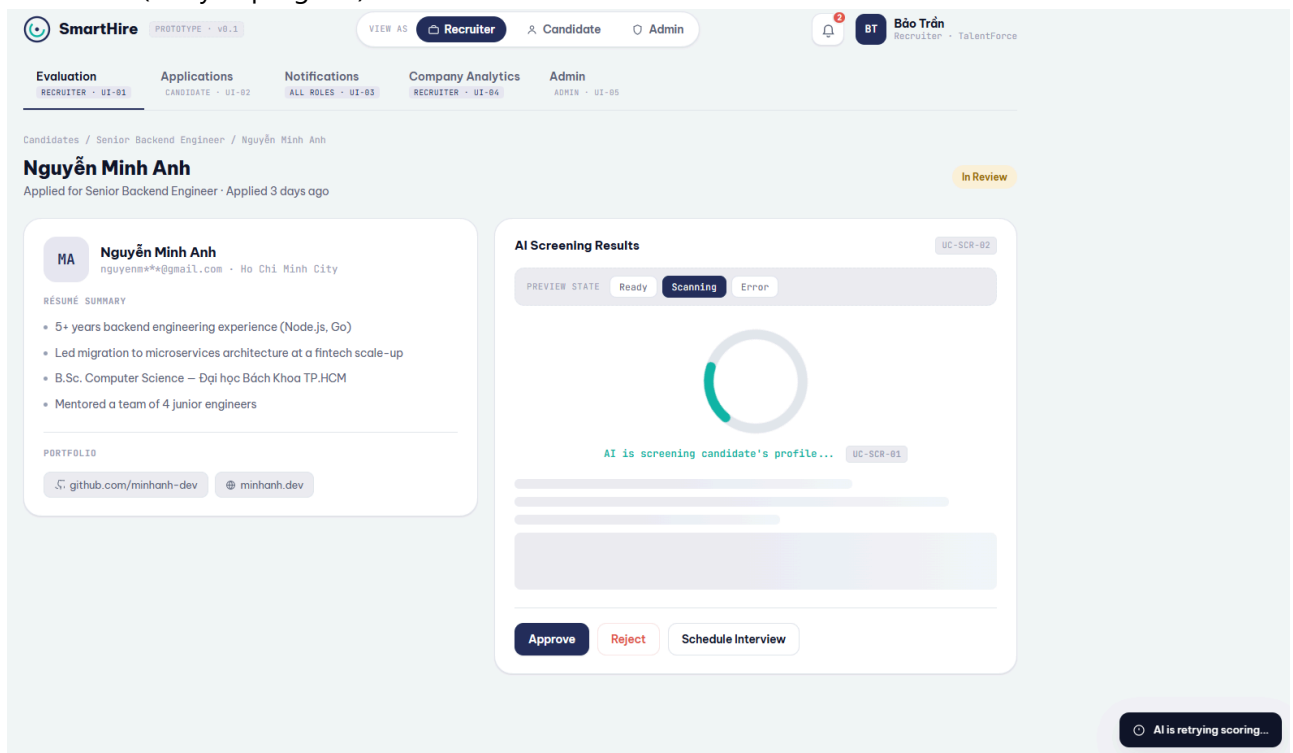
- **Use-case ID:** UC-SCR-04
- **Use-case Name:** Retry Failed Scoring
- **Actor(s):** Recruiter (Authorized), System
- **Description:** Allows a recruiter or the system to retry the hybrid scoring process for an application that previously encountered an AI service failure, without changing the application's overall recruitment stage.
- **Preconditions:** The application has a screening status of **Failed**.
- **Basic Flow:**
 1. **Recruiter** views an application with a failed scoring status and clicks "Retry AI Scoring".
 2. **System** confirms the action and changes the status back to **Processing**.
 3. **System** re-triggers UC-SCR-01.
 4. **System** displays a progress toast notification to the user.
- **Alternative Flows:**
 - **AF-01: Retry Fails Again (at Step 3):**
 1. If the system fails again, it reverts the status to **Failed** and notifies the recruiter.
- **Postconditions:** The screening process is successfully restarted.
- **Special Requirements:** None.

Prototype Evidence:

- Basic Flow (Error visible):



- Basic Flow (Retry in progress):



4.4. UC-NOT-01: Receive Event Notification

- **Use-case ID:** UC-NOT-01
- **Use-case Name:** Receive Event Notification
- **Actor(s):** Candidate, Company Member (Authorized)
- **Description:** Delivers system-generated notifications (in-app alerts and emails) to relevant actors when recruitment, verification, or account events occur.
- **Preconditions:** A trigger event has occurred (e.g., application stage change, moderation decision).

- **Basic Flow:**

1. **System** generates a notification payload based on the event.
2. **System** delivers an in-app alert (e.g., toast notification or dropdown update) to the active session of the recipient.
3. **Actor** views the newly surfaced alert indicator on their interface.

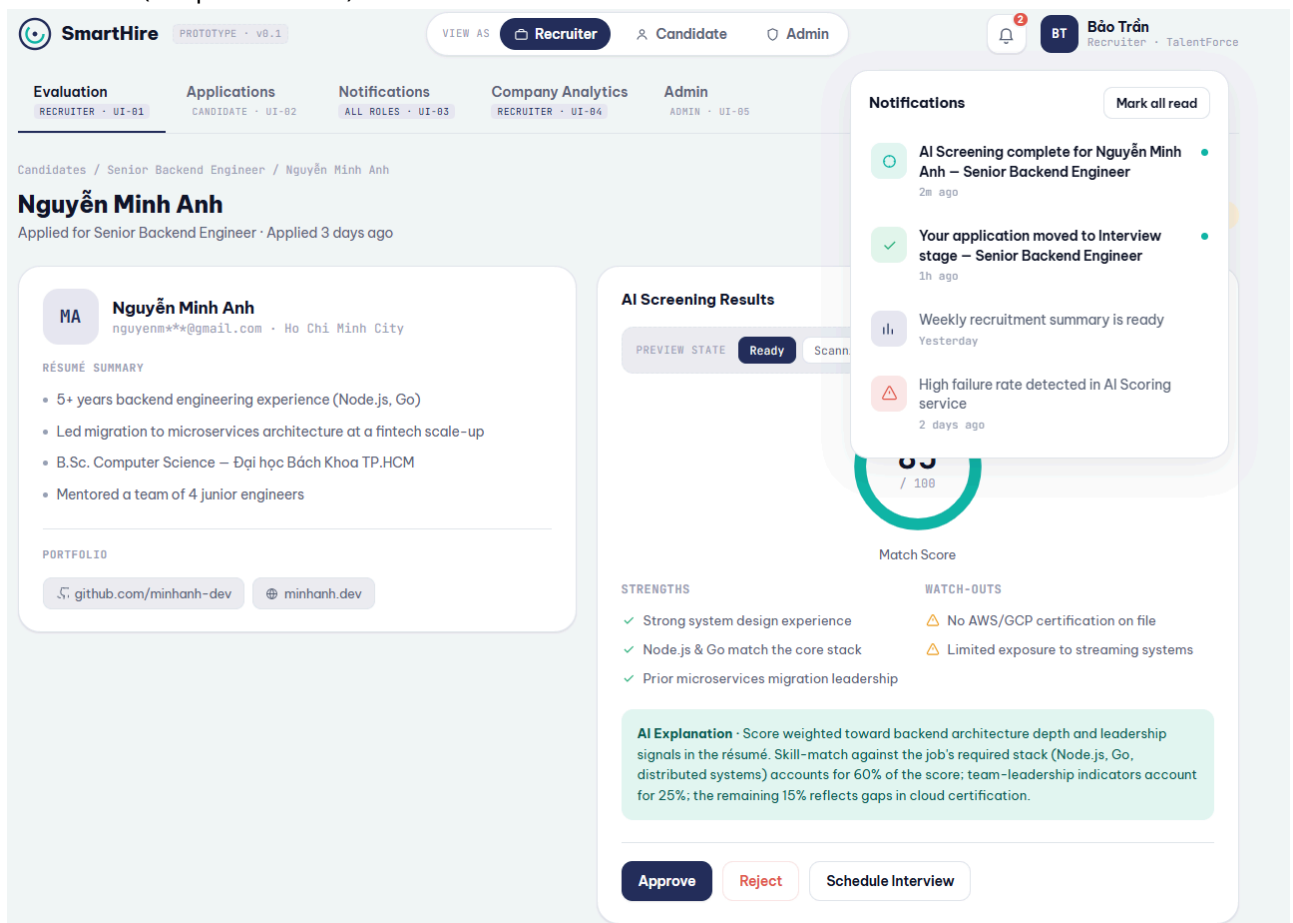
- **Alternative Flows:** None.

- **Postconditions:** The notification is delivered and added to the user's unread list.

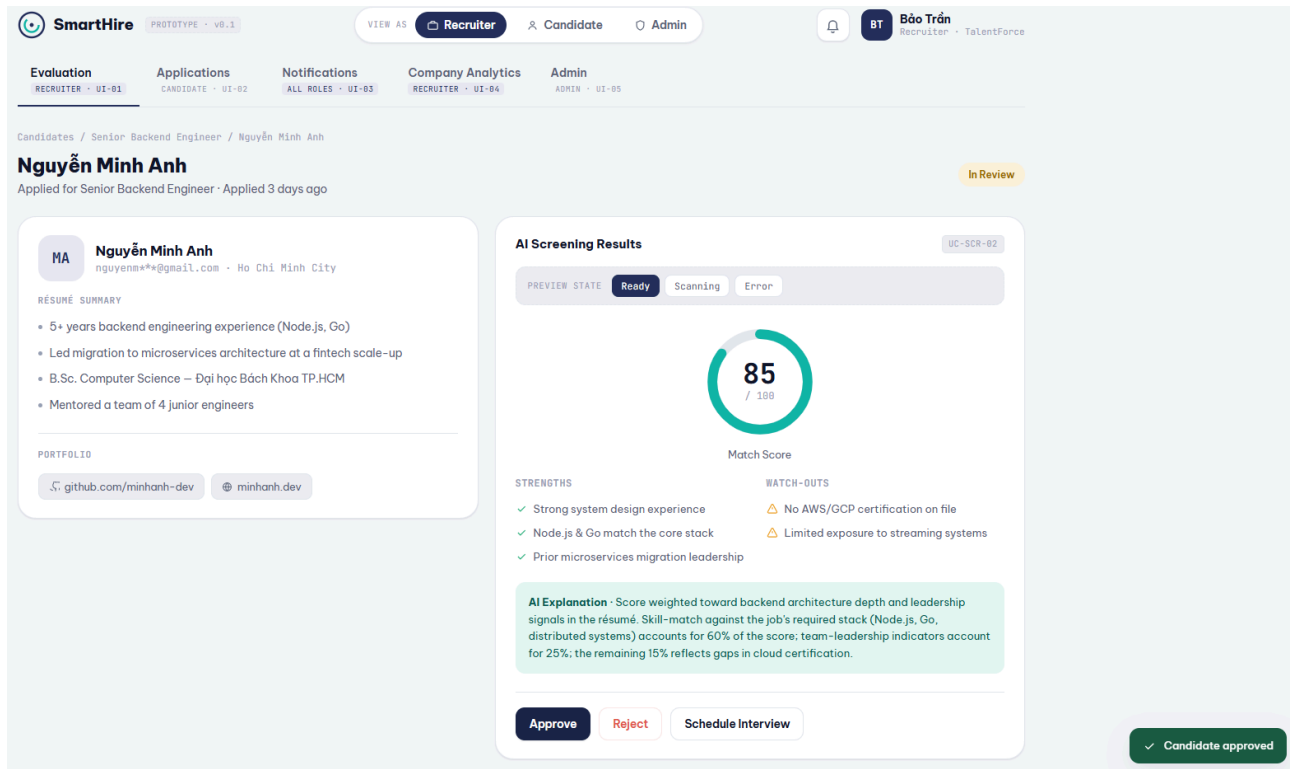
- **Special Requirements:** In-app notifications must be delivered to active user sessions in near real-time (e.g., via WebSocket).

Prototype Evidence:

- Basic Flow (Dropdown Alert):



- Basic Flow (Toast Alert):

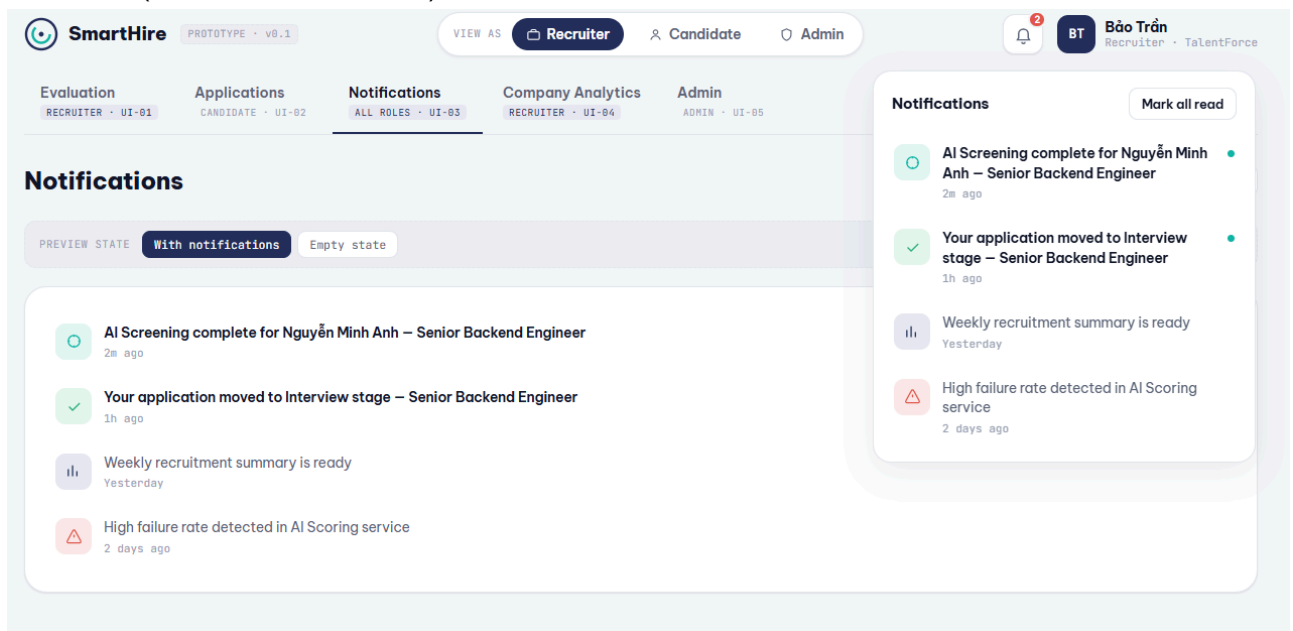


4.5. UC-NOT-02: Manage In-App Notifications

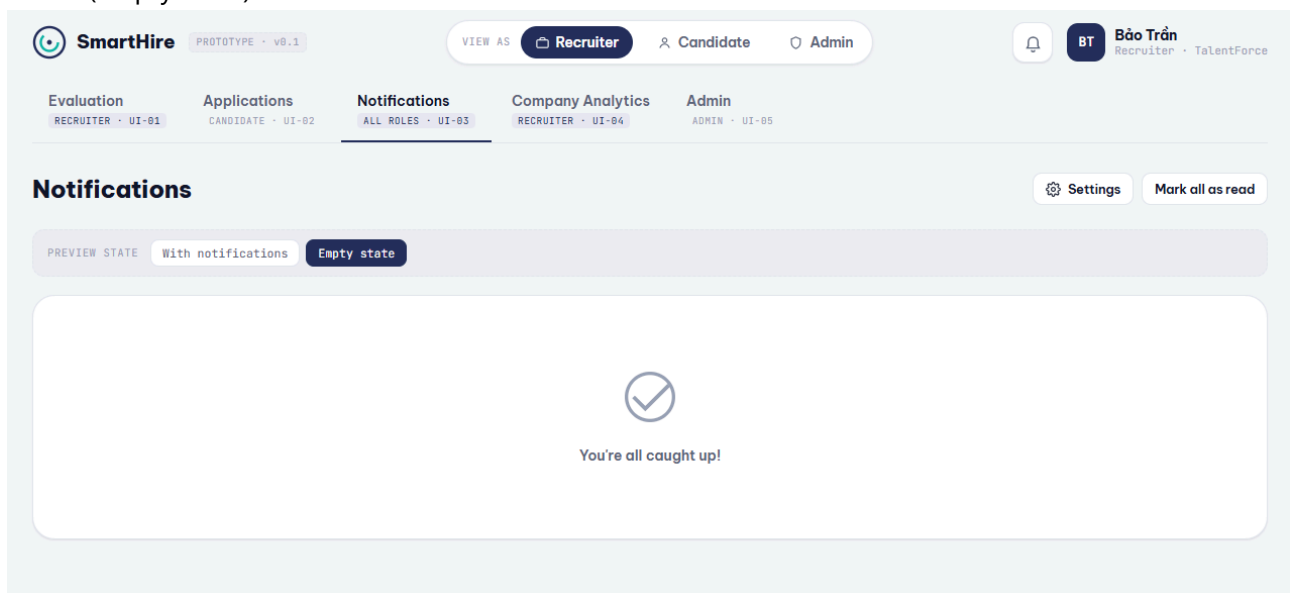
- **Use-case ID:** UC-NOT-02
- **Use-case Name:** Manage In-App Notifications
- **Actor(s):** Authenticated User
- **Description:** Allows users to view their notification list, distinguish between read and unread items, and mark notifications as read.
- **Preconditions:** The user is logged in.
- **Basic Flow:**
 1. **Authenticated User** navigates to the Notifications tab.
 2. **System** retrieves the user's notification history.
 3. **System** displays the list, highlighting unread notifications with a visual indicator.
 4. **User** clicks "Mark all as read".
 5. **System** updates the status of all notifications to read and removes the visual indicators.
- **Alternative Flows:**
 - **AF-01: Empty State (at Step 2):**
 1. If the user has no notifications, the **System** displays an empty state message ("You're all caught up!").
- **Postconditions:** The notification statuses are updated in the database.
- **Special Requirements:** None.

Prototype Evidence:

- Basic Flow (List with Unread status):



- AF-01 (Empty State):



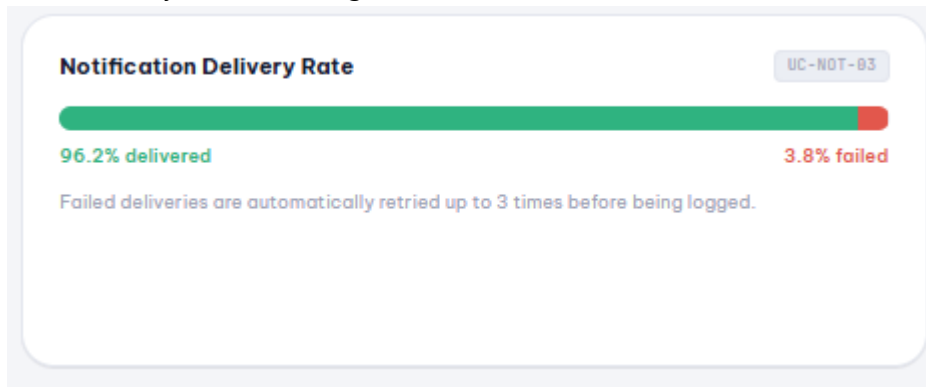
4.6. UC-NOT-03: Retry Failed Notification Delivery

- **Use-case ID:** UC-NOT-03
- **Use-case Name:** Retry Failed Notification Delivery
- **Actor(s):** System
- **Description:** The system automatically logs failed external notification attempts (e.g., email delivery failures) and retries them according to a configured schedule.
- **Preconditions:** A notification delivery attempt has failed.
- **Basic Flow:**
 1. **System** detects a failed delivery attempt to an external channel.
 2. **System** logs the failure and increments the retry counter.
 3. **System** pauses for the configured delay, then re-attempts delivery.
 4. **System** updates aggregate analytics regarding delivery success/failure rates.
- **Alternative Flows:** None (internal system process).

- **Postconditions:** The notification is either delivered successfully or logged as permanently failed after max retries.
- **Special Requirements:** Retry mechanism should implement exponential backoff to avoid overloading external email services.

Prototype Evidence:

- Basic Flow (System recording failure metrics):

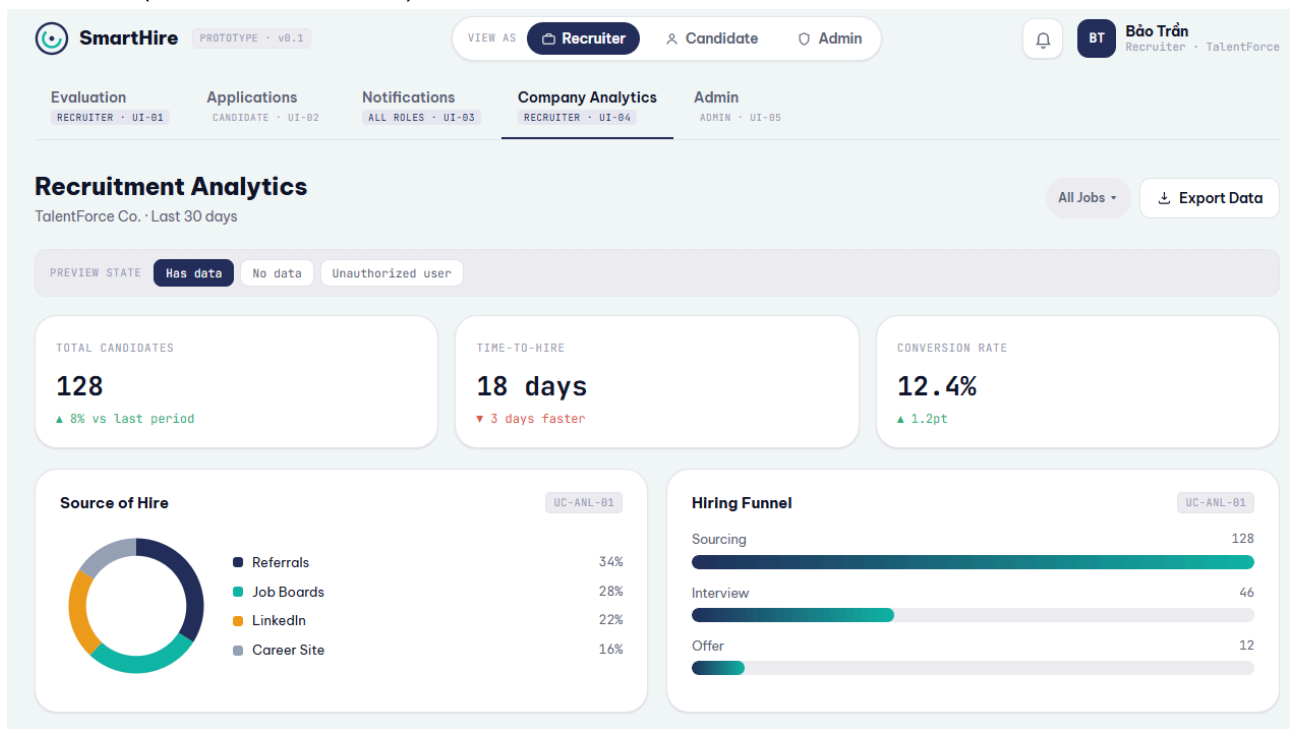


4.7. UC-ANL-01: View Company Recruitment Analytics

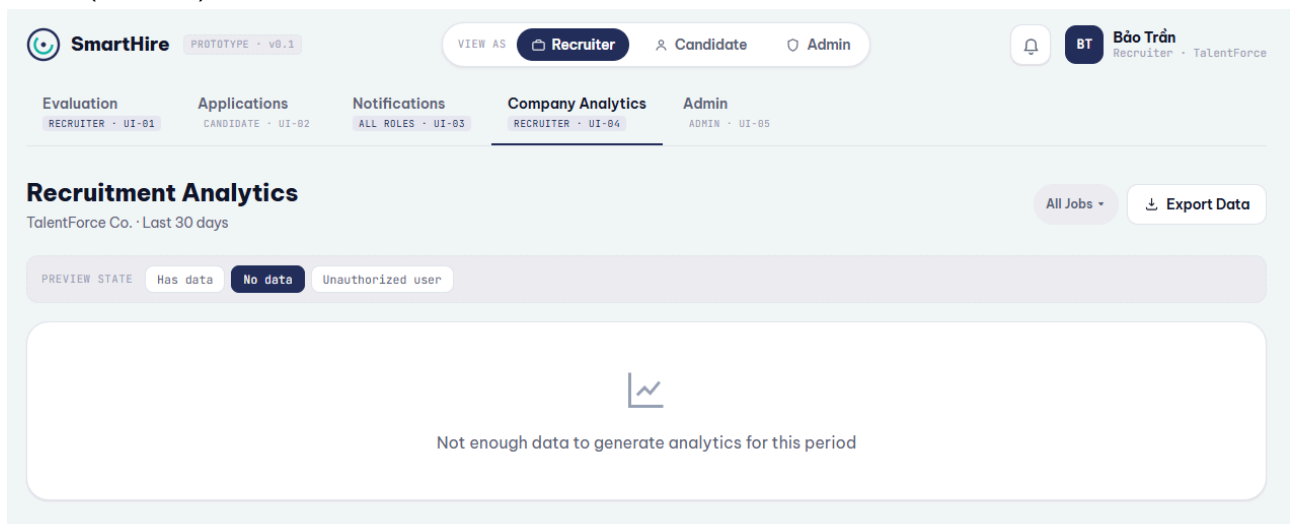
- **Use-case ID:** UC-ANL-01
- **Use-case Name:** View Company Recruitment Analytics
- **Actor(s):** Company Member (Authorized)
- **Description:** Allows an authorized company member to view aggregated recruitment metrics (e.g., time-to-hire, source of hire, hiring funnel) scoped strictly to their active company.
- **Preconditions:** The user is authenticated and holds the required company-scoped permissions.
- **Basic Flow:**
 1. **Company Member** navigates to the Company Analytics dashboard.
 2. **System** validates the user's company membership role.
 3. **System** retrieves aggregated recruitment data for the active company context.
 4. **System** renders the metrics, charts (donut, funnel), and trends on the dashboard.
- **Alternative Flows:**
 - **AF-01: Insufficient Data (at Step 3):**
 1. If there is not enough historical data to generate meaningful analytics, the **System** displays a "Not enough data" empty state.
 - **AF-02: Permission Denied (at Step 2):**
 1. If the user lacks the required role, the **System** disables analytical features and denies access to restricted actions.
- **Postconditions:** The user views the dashboard populated with their company's data.
- **Special Requirements:** None.

Prototype Evidence:

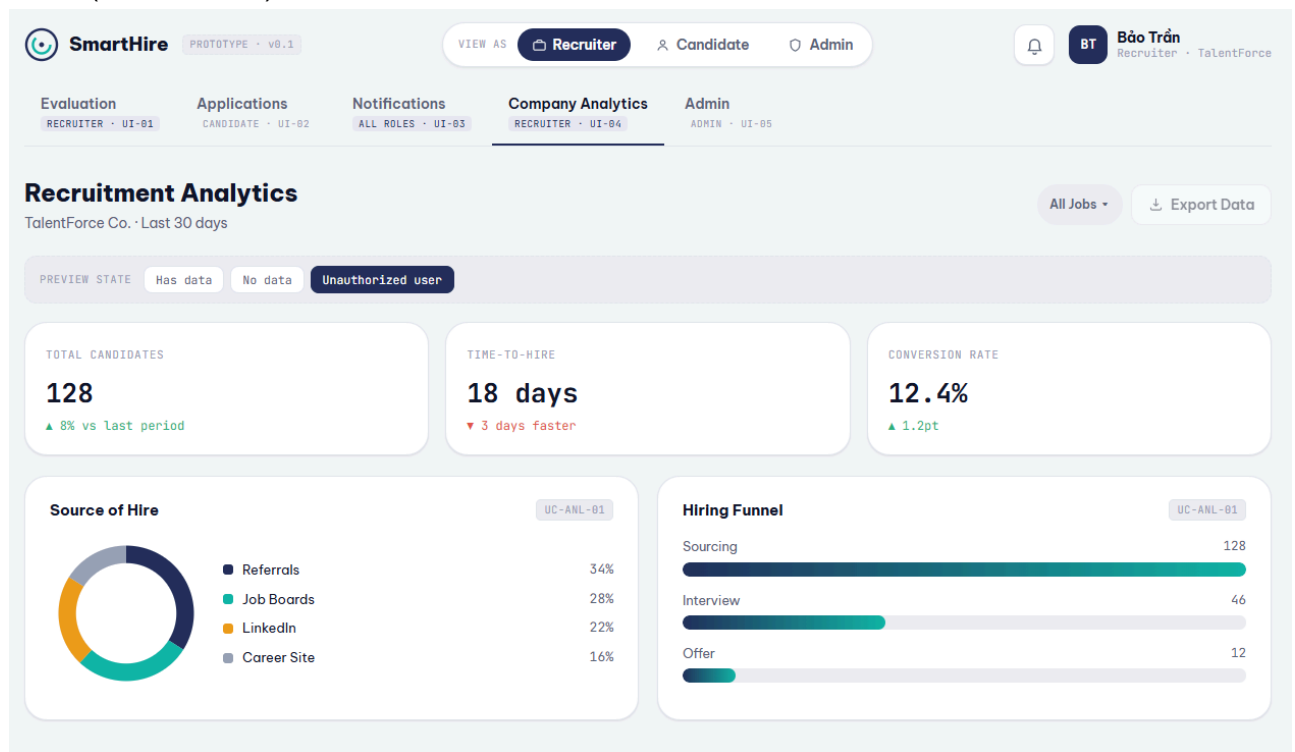
- Basic Flow (Dashboard with data):



- AF-01 (No Data):



- AF-02 (Unauthorized):

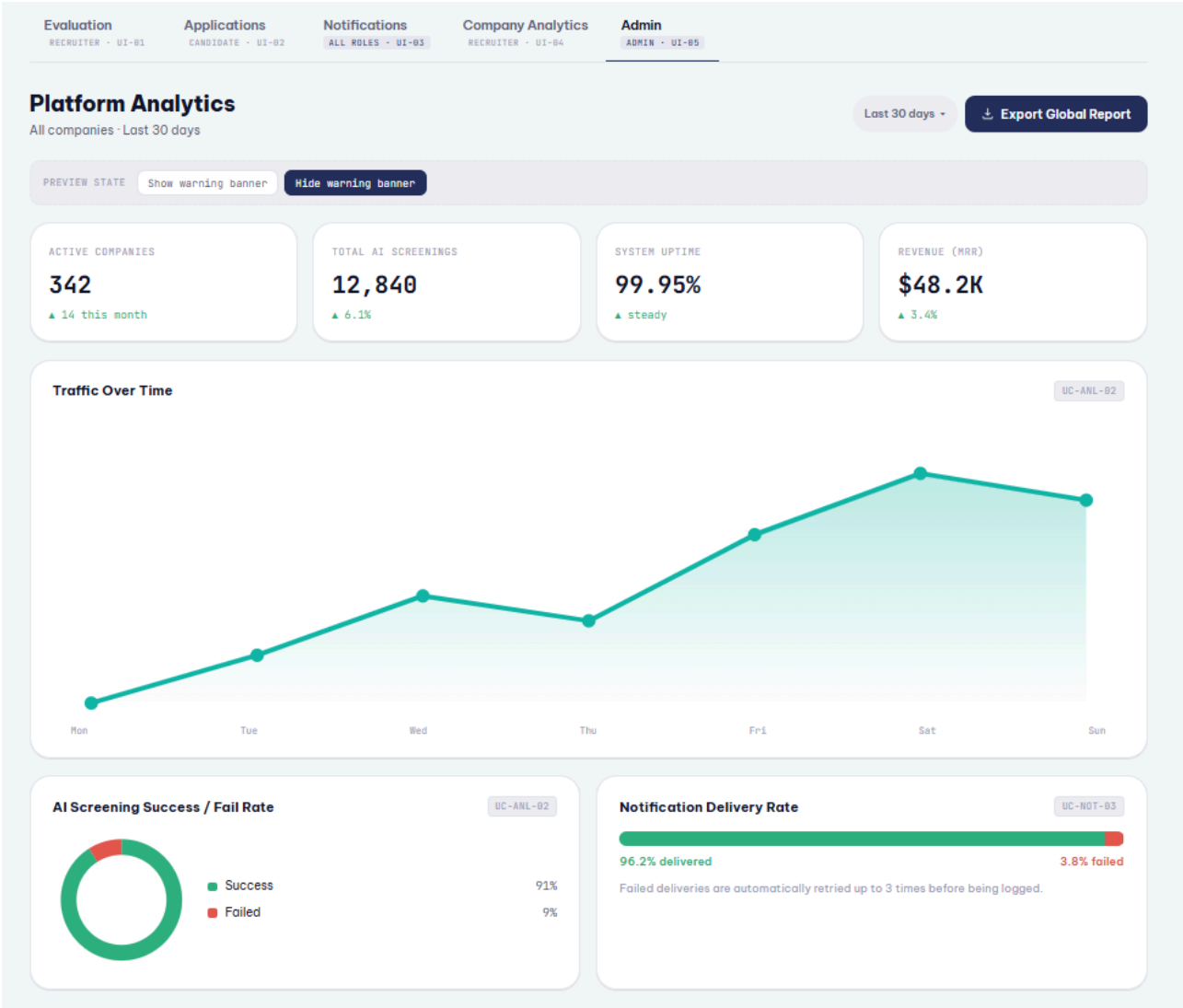


4.8. UC-ANL-02: View Platform Analytics

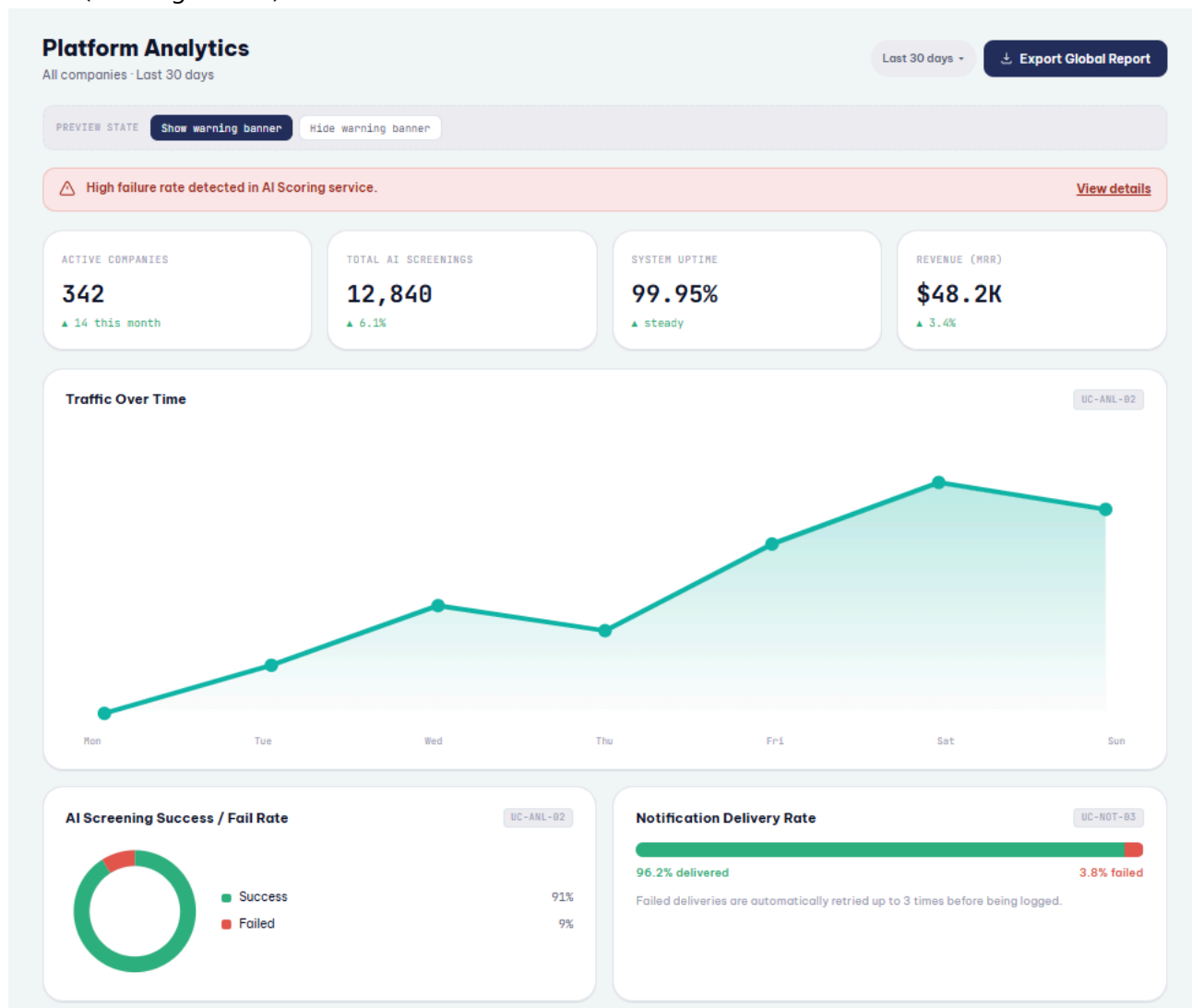
- **Use-case ID:** UC-ANL-02
- **Use-case Name:** View Platform Analytics
- **Actor(s):** Platform Administrator
- **Description:** Allows a Platform Administrator to view global, platform-wide metrics including active companies, total AI screenings, system uptime, and aggregate revenue.
- **Preconditions:** The user is authenticated as a Platform Administrator.
- **Basic Flow:**
 1. **Platform Administrator** navigates to the Admin Dashboard.
 2. **System** verifies the administrator role.
 3. **System** aggregates global platform data.
 4. **System** displays traffic line charts, success/failure rate donuts, and top-level metric cards.
- **Alternative Flows:**
 - **AF-01: System Anomaly Detected (at Step 4):**
 1. If the **System** detects abnormal failure rates (e.g., AI Scoring service disruption), it renders a high-priority warning banner at the top of the dashboard.
- **Postconditions:** The administrator receives a comprehensive view of platform health and statistics.
- **Special Requirements:** None.

Prototype Evidence:

- Basic Flow (Admin Dashboard):



- AF-01 (Warning Banner):



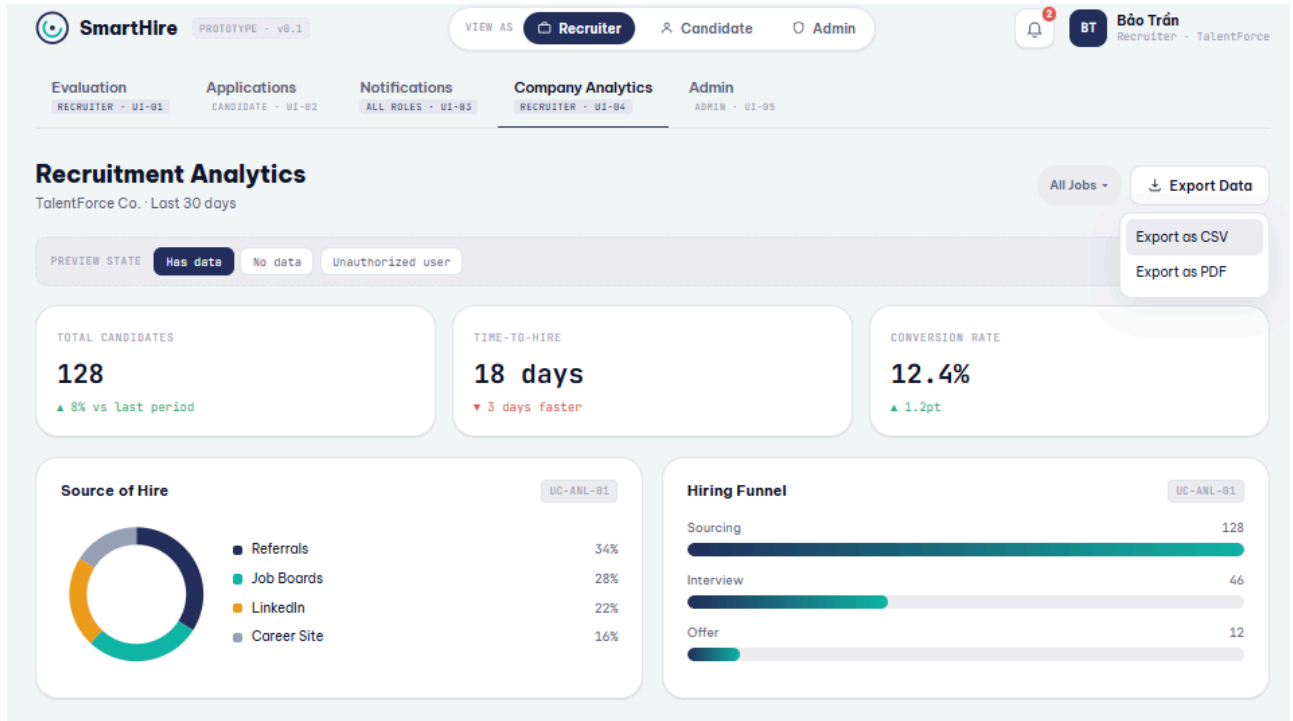
4.9. UC-ANL-03: Export Authorized Data

- **Use-case ID:** UC-ANL-03
- **Use-case Name:** Export Authorized Data
- **Actor(s):** Authorized User(Company Member, Platform Administrator)
- **Description:** Allows a user with the appropriate permissions to export filtered recruitment analytics or platform data into external formats (CSV or PDF).
- **Preconditions:** The user is viewing an analytics dashboard and has export privileges.
- **Basic Flow:**
 1. **Authorized User(Company Member, Platform Administrator)** clicks the "Export Data" button on the dashboard.
 2. **System** displays a dropdown menu with available format options (CSV, PDF).
 3. **User** selects a format (e.g., CSV).
 4. **System** compiles the authorized data payload and triggers the file download.
 5. **System** displays a success toast notification.
- **Alternative Flows:**
 - **AF-01: Permission Denied (at Step 1):**
 1. If the **System** determines the user lacks export privileges for the current context, the "Export Data" button is disabled and blocked from interaction.

- **Postconditions:** A file containing the requested data is downloaded to the user's device.
- **Special Requirements:** Exported documents (especially PDF) must adhere to the system's data privacy policies, ensuring sensitive PII is masked if the user's role requires it.

Prototype Evidence:

- Basic Flow (Format Menu):



- AF-01 (Disabled Button):

